

TIMEMIXER++: A General Time Series Pattern Machine for Universal Predictive Analysis

Shiyu Wang, Jiawei Li, Xiaoming Shi, Zhou Ye, Baichuan Mo, Wenze Lin, Shengtong Ju, Zhixuan Chu, Ming Jin

01 TIMEMIXER++: Title and Authorship

Authors: Shiyu Wang, Jiawei Li, Xiaoming Shi, Zhou Ye, Baichuan Mo, Wenze Lin, Shengtong Ju, Zhixuan Chu, Ming Jin. Affiliations: Griffith University; HKUST Guangzhou; MIT; Zhejiang University; The State Key Laboratory of Blockchain and Data Security; Hangzhou High-Tech Zone Institute of Blockchain and Data Security. Emails: kwuking@gmail.com, jarvis-li@outlook.com, sxm728@hotmail.com, yezhou199032@gmail.com, baichuan@mit.edu, zhixuanchu@zju.edu.cn, {linwenze75, shengtongju, mingjinedu}@gmail.com

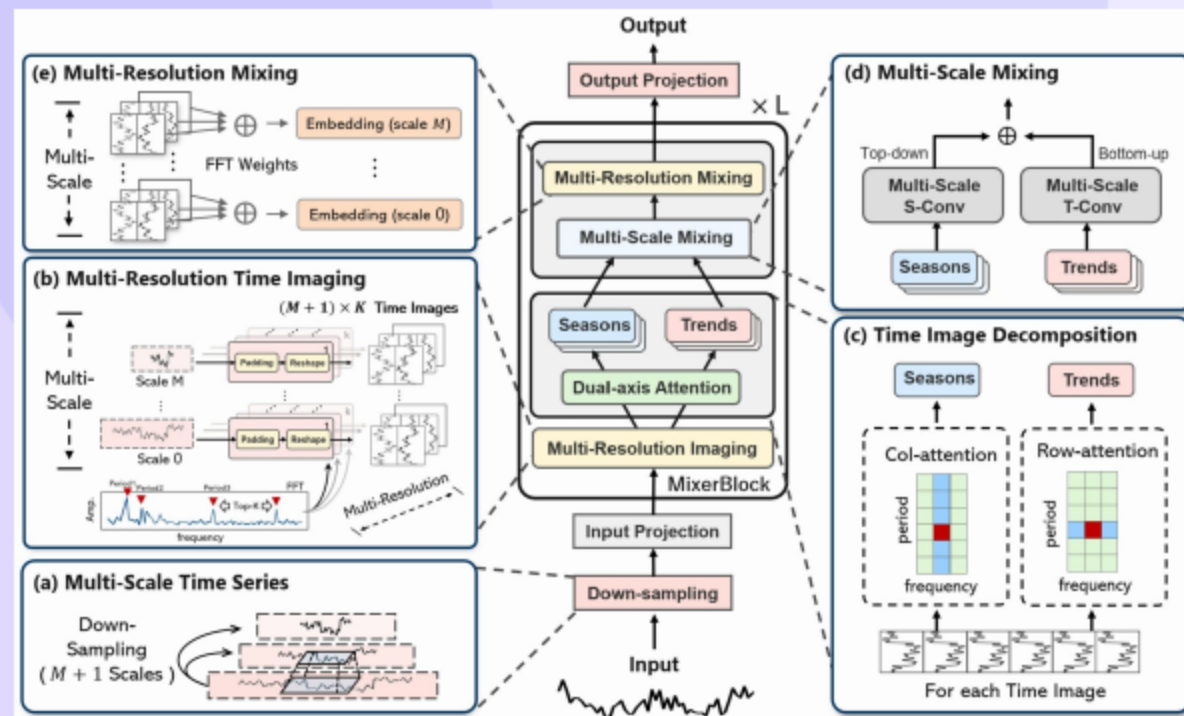
02 Motivation: Challenges in Universal TS Modeling

- Universal time series modeling spans forecasting, imputation, classification, and anomaly detection
- Recurrent and convolutional models struggle with long dependencies while attention helps long-range context
- A single robust model must remain effective across tasks, horizons, and scales despite noise and sparsity

03 Motivation (cont.): Overlapping Periodicities and Tokenization Gaps

- Overlapping daily, weekly, and seasonal patterns blur token boundaries for sequence models
- Multi-scale and multi-periodicity dynamics demand adaptive pattern extraction without brittle tokenization
- Effective representations should align recurring motifs while tracking evolving trends over time

04 Problem Framing: What a General TSPM Must Do



- A general pattern machine captures local motifs and long-range seasonalities without fixed windows
- Frequency-aware adaptation discovers overlapping and time-varying periodicities across contexts
- Cross-variable modeling learns shared structure and idiosyncratic behaviors in multivariate settings
- Task-adaptive representations support forecasting, imputation, classification, and anomaly objectives
- Robustness to sparsity, noise, and shift ensures stable performance under non-stationarity
- Efficiency and interpretability enable scalable learning with clear seasonal and trend views

05 Prior Work: Sequence Models for TS (RNN/TCN/Transformer)

- Classical models offer structured seasonality and trend but lack nonlinear flexibility
- Recurrent and convolutional baselines face limits on long horizons and global context
- Transformer and simple MLP variants show **strong long-range performance** in recent studies

06 Prior Work: Hierarchical/Decomposition Approaches

- Rigid predefined decompositions limit adaptability to complex and drifting dynamics
- Latent-space disentanglement with dual-axis attention enables flexible seasonal and trend views
- Hierarchical cross-scale interaction improves coverage of multi-periodic structure

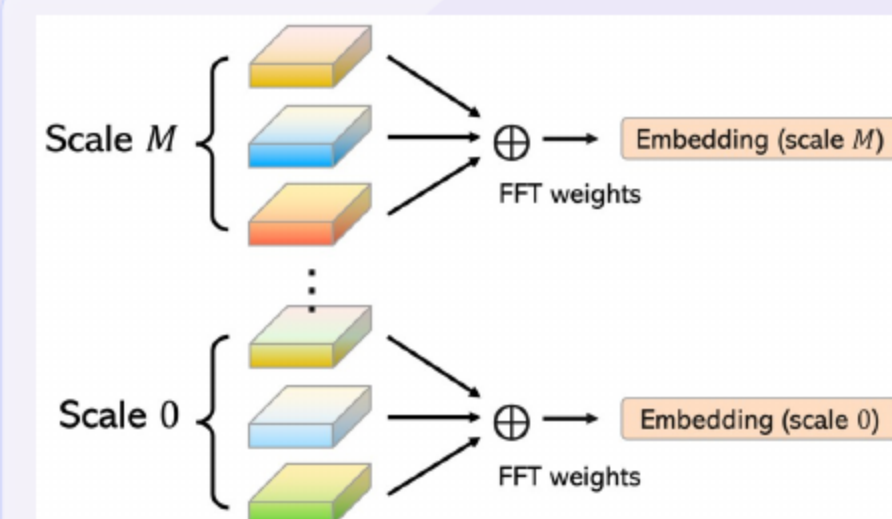
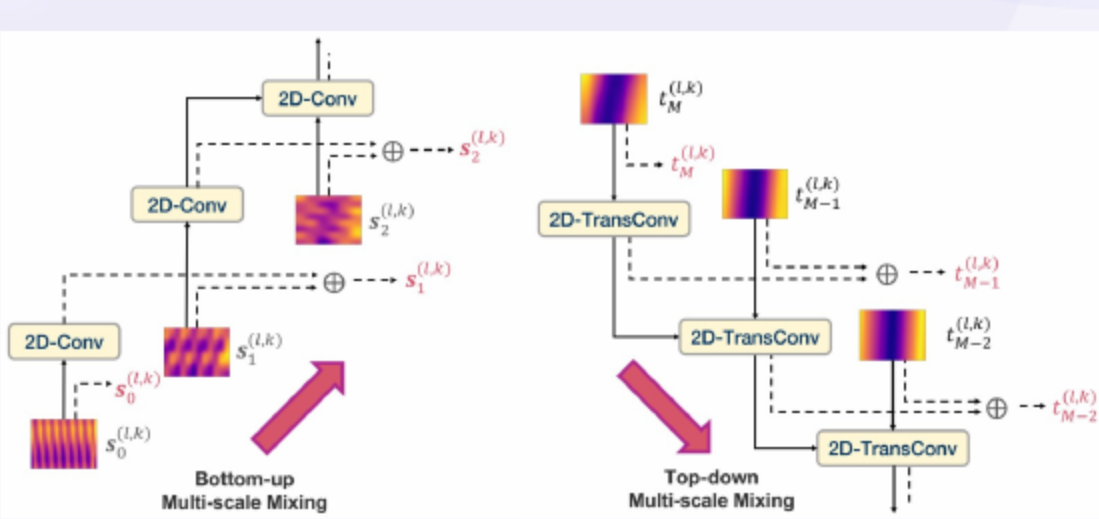
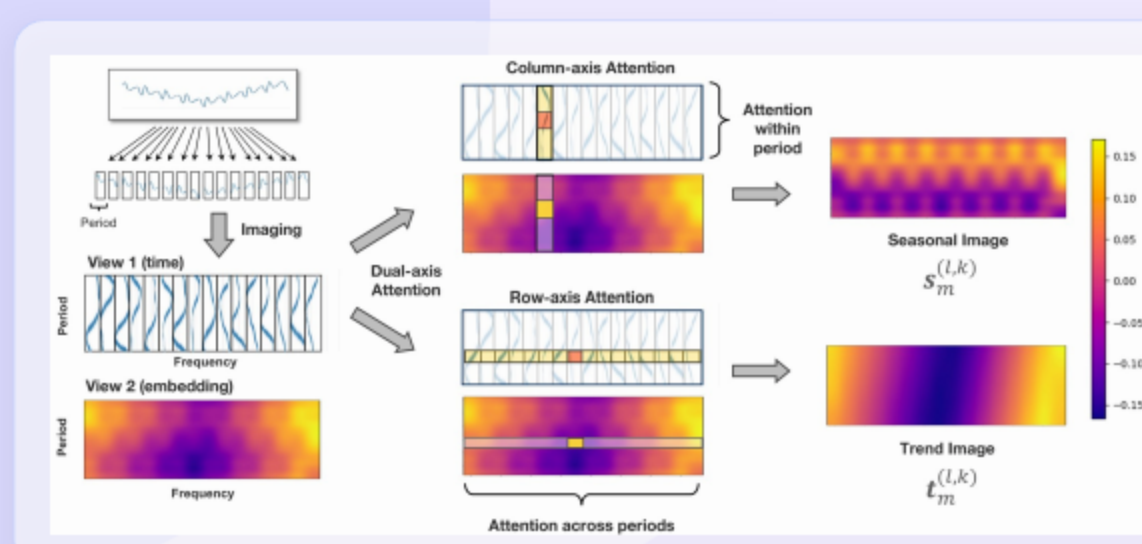
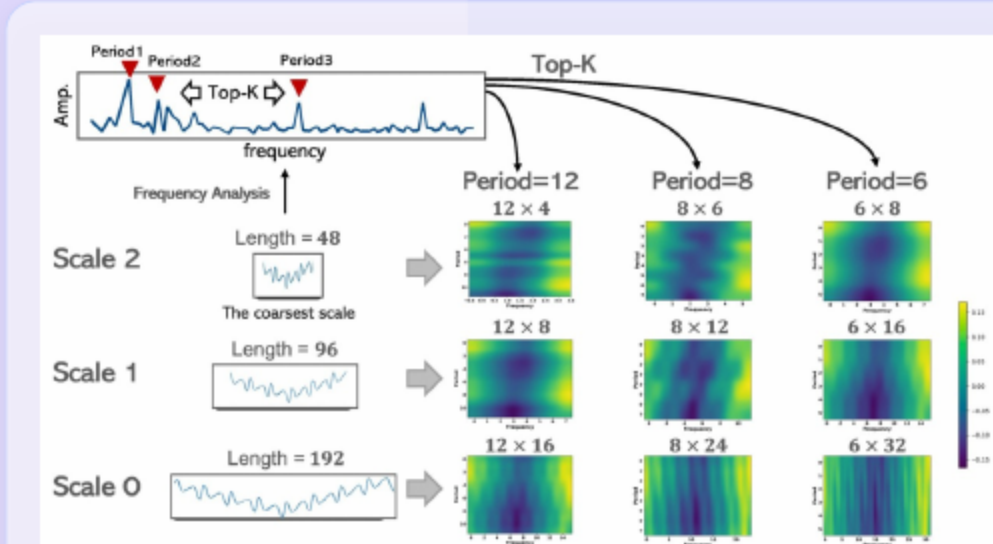
07 Related SOTA at a Glance

- Strong forecasting baselines include iTransformer, PatchTST, TimesNet, TiDE, and TimeMixer
- Imputation and anomaly tasks often reuse forecasting backbones with masking or detection heads
- Traffic and high-dimensional sensors benchmark multivariate scalability and cross-variable coupling

- Multi-resolution time imaging, dual-axis decomposition, and hierarchical mixing form a unified design
- They report [consistent state-of-the-art](#) across diverse tasks
- They show task-dependent representation profiles that align shared structure with specialized features

- An encoder-only stack projects inputs, processes with MixerBlocks, and aggregates with scale heads
- Channel mixing captures cross-variable dependencies before multi-scale embedding
- Scale-specific predictions are ensembled to match target horizons and tasks

10 Inside MixerBlock: MRTI → TID → MCM → MRM



11 Ablation Study — why components matter

	ETTh1	ETTh2	ETTm1	ETTm2	ECL	Traffic	Weather	Solar	Average	Promotion
TIMEMIXER++	0.419	0.339	0.369	0.269	0.165	0.416	0.226	0.203	0.300	-
w/o channel mixing	0.424	0.346	0.374	0.271	0.197	0.442	0.233	0.245	0.317	5.36%
w/o time image decomposition	0.441	0.358	0.409	0.291	0.198	0.445	0.251	0.241	0.329	8.81%
w/o multi-scale mixing	0.447	0.361	0.391	0.284	0.172	0.427	0.239	0.234	0.320	6.25%
w/o multi-resolution mixing	0.431	0.350	0.374	0.280	0.181	0.432	0.241	0.233	0.316	5.10%

- Removing time image decomposition, they observe **large degradation** on overlapping and drifting periodicities
- Eliminating multi-scale mixing, they observe weakened long-horizon accuracy due to reduced cross-scale flow
- Disabling multi-resolution fusion and channel mixing, they report reduced stability and multivariate gains

12

Representation Analysis — what the model learns

- Seasonal and trend paths remain distinct while aligning motifs and tracking baselines
- Representation similarity profiles vary by task to balance shared and specialized features
- Task adaptive structure supports better generalization across objectives and domains

13

Results Across 8 Tasks (brief setup)

Models	TimeMixer++ (Ours)		TimeMixer (2024b)		iTransformer (2024)		PatchTST (2023)		Crossformer (2023)		TiDE (2023a)		TimesNet (2023)		DLinear (2023)		SCINet (2022a)		FEDformer (2022b)		Stationary (2022c)		Autoformer (2021)	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Electricity	0.165	0.253	0.182	0.272	0.178	0.270	0.205	0.290	0.244	0.334	0.251	0.344	0.192	0.295	0.212	0.300	0.268	0.365	0.214	0.327	0.193	0.296	0.227	0.338
ETT (Avg)	0.349	0.399	0.367	0.388	0.383	0.377	0.381	0.397	0.685	0.578	0.482	0.470	0.391	0.404	0.442	0.444	0.689	0.597	0.408	0.428	0.471	0.464	0.465	0.459
Exchange	0.357	0.391	0.391	0.453	0.378	0.360	0.403	0.404	0.940	0.707	0.370	0.413	0.416	0.443	0.354	0.414	0.750	0.626	0.519	0.429	0.461	0.454	0.613	0.539
Traffic	0.416	0.264	0.484	0.297	0.428	0.282	0.481	0.304	0.550	0.304	0.760	0.473	0.620	0.336	0.625	0.383	0.804	0.509	0.610	0.376	0.624	0.340	0.628	0.379
Weather	0.226	0.262	0.240	0.271	0.258	0.278	0.259	0.281	0.259	0.315	0.271	0.320	0.259	0.287	0.265	0.317	0.292	0.363	0.309	0.360	0.288	0.314	0.338	0.382
Solar-Energy	0.203	0.238	0.216	0.280	0.233	0.262	0.270	0.307	0.641	0.639	0.347	0.417	0.301	0.319	0.330	0.401	0.282	0.375	0.291	0.381	0.261	0.381	0.885	0.711

- They evaluate across 8 tasks and 30 benchmarks
- They observe consistent improvements over 27 baselines

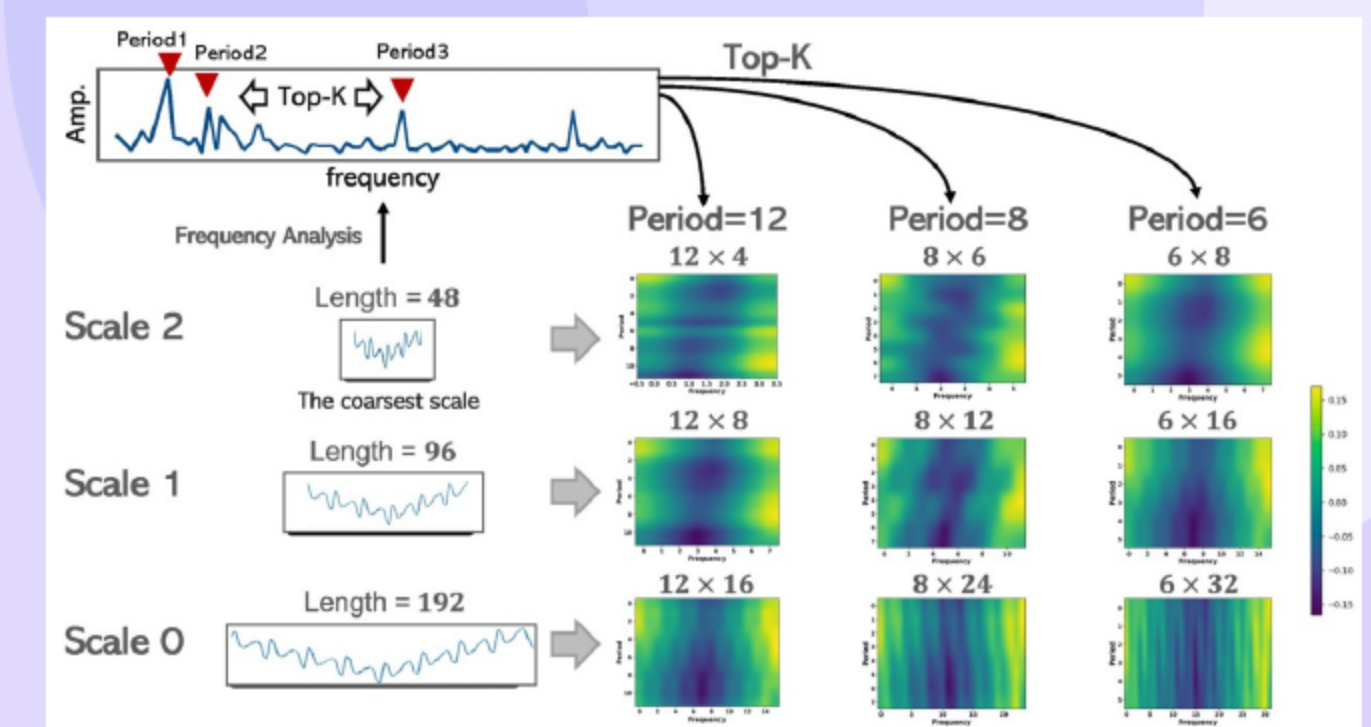
14

From Motivation to Practice: When and Why TIMEMIXER++ Helps

- These settings benefit; they show that time images and dual-axis decomposition help with multi-periodic, cross-scale dynamics
- These settings benefit; they show gains on high-dimensional sensors from cross-variable channel mixing and scale ensembles
- These settings benefit; they show improvements in sparse data and transfer with adjustable frequency count and capacity placement

15

Main Failure Mode: Sensitivity to Frequency Selection and Weak Periodicity



- Top period selection is sensitive to noisy spectra and mis specified counts during imaging
 - Mitigate with learned frequency gating and broader candidate pools
 - Stabilize with spectral jitter and periodic re estimation
- Weak or non stationary seasonality undermines frequency derived representations
 - Fallback to time domain paths when confidence is low
 - Use change point cues to damp frequency contributions

- The approach is **strong overall** yet remains sensitive to frequency selection in challenging regimes
- They propose future work including adaptive frequency discovery and richer time-domain fallbacks
- They plan broader benchmarks and deployment studies to validate scalability and robustness

-
- They deliver clear seasonal and trend decomposition via time images and dual-axis attention
 - They produce **robust performance** across tasks through cross-scale and cross-resolution mixing
 - They support efficient and transferable modeling with a unified encoder and scale heads
-

Thank You

Questions & Discussion